

Generative AI: Algorithms and Apps, Summer 2026

Project: Diffusion

0.1 Problem Description

You will be implementing a DDPM on MNIST dataset, with and without classifier free guidance. Before starting, make sure you have a good understanding of the class notes. All notations are from this [tutorial](#). We follow the epsilon-estimation loss formulation. We have provided a diffusion denoising U-Net, a diffusion scheduler and helper functions, visualization code for conditional and unconditional generation.

0.1.1 Part A: Complete the Missing Loss Functions

Some functions/modules in the starter code (`cfg_starter.ipynb`) are incomplete. You are required to complete those functions. We will test the accuracy of the implementation with custom inputs.

You are required to complete the following functions:

1. DDPM epsilon estimation loss (`ddpm_loss_epsilon()`).
2. DDPM epsilon estimation loss for classifier free guidance (`ddpm_loss_cfg()`).

Gradescope will only check the correctness of the loss functions.

0.1.2 Part B: Sampling and Experiments

Using the completed code from Part A, follow the instructions in the notebook and do the following:

1. Train the diffusion model following the notebook configuration.
2. Complete the code of DDPM unconditional sampling (`sample_unconditional()`)
3. Complete the code of DDPM sampling with classifier free guidance (`sample_conditional()`)
4. Use your unconditional and conditional sampling code to sample images, visualize using the given code
5. Run ablations on conditional generation with $\gamma \in \{-1, 0, 1, 2, 3, 4, 5\}$. Explain what different γ means while explaining the generated images.

0.2 Exploration

1. **Can you generate a similar image by sampling nearby latent codes?:** Say you generated an image of digit "7" using the diffusion model. Now, you noisify the image to certain diffusion step t and slightly perturb it by adding a small noise. Now, you denoise the perturbed noisy image back to $t = 0$. What do you observe? Is your model able to generate a similar image? If yes, why? If not, why not? Does your answer change with different t or use of unconditional/conditional sampling?
2. **Can diffusion fill holes?:** Say you generated an image of digit "8" using the diffusion model. Now, you mask a part of the image (say the top part of the digit) and noisify the image to certain diffusion step t . Now, you denoise the masked noisy image back to $t = 0$. What do you observe? Is your model able to fill the hole? If yes, why? If not, why not? Does your answer change with different t or use of unconditional/conditional sampling?